An Explainable Machine Learning Model for Assessing Social Media Addiction Among Students

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Abstract--Social media addiction in students has recently become an issue of increasing concern with its adverse effect on academic achievements, mental health, and general lifestyle. Therefore, the possibility to study and identify the trends of overuse has become a proactive field of research that helps to detect vulnerable groups automatically. The features are significant in the social media addiction system since they define the nature of the digital behaviour of the students. We suggest a complex model of the user behaviou in this paper, based on the variables of general frequency of use, the most frequently used platforms, the number of hours of sleep, the degree of addiction, and the elements of mental health. In all, the relationship between usage habits and academic or psychological effects is investigated by deriving and feeding five important behavioral aspects to predictive models. To evaluate its performance, a sample of 705 students is taken into consideration and the results are evaluated using the descriptive statistics and correlation based measures.

Keywords--Online Social Media, Rumours, user's behaviour, author, readers, Classifier, Precision.

I. INTRODUCTION

The advent of the Online Social Media (OSM) platform which includes Facebook, Instagram, WhatsApp, and YouTube has recently gained a lot of popularity, especially among students, since they serve as sources of entertainment, communication, and rapid access to information. Students are also able to exchange content with each other and keep up with the global trends with the assistance of OSM. But the growing reliance on these platforms has had an effect. felt the need to address the issue of social media addiction and its negative impact on the academic performance of students, their health, and personal lifestyle [1].Although social media has the positive consequences of the potential to share knowledge and network, research has demonstrated the appearance of addiction-like symptoms such as compulsive checking, self-control loss, and failure to fulfill duties along with other life obligations [2]. This behavior has been linked to low performance in schools, anxiety, depression, and disturbed sleeping habits. Indicatively, students who use social media late into the night tend to complain of reduced sleep quality, exhaustion and loss of concentration in the classroom tasks [3]. Social media addiction is therefore emerging to be a socio-psychological challenge in learning institutions. As a rule, social media addiction identification can be addressed as a prediction and classification issue. The student can be divided into various levels of addiction depending on their demographic characteristics, hours spent on daily use, platform, amount of sleep, and addiction level [4]. selection of features like lifestyle predictors, hours of study, and frequency of conflicts, which have an overwhelming impact on addiction tendencies [6]. To curb this, the current study suggests an attribute-based process, i.e. feature-driven, which incorporates behavioral characteristics of the students into predictive models. In the work, we preprocess the data on social media usage of students and come up with meaningful features that reflect their digital way of life. These characteristics are then trained on machine learning classifiers such as Logistic Regression, Decision Trees and Random Forests, to make predictions about the levels of social media addiction. The main goal will be to analyze the ways in which behavioral and lifestyle patterns. influence dependency and to determine at-risk students. The remaining sections of the paper are structured in the following way: Section II presents the associated literature review of social media addiction. Section III presents the suggested methodology and feature extraction procedure. The discussion and research results are presented in section IV. Part V is a conclusion of the research with major conclusions and scope.

II. LITERATURE SURVEY

Kuss and Griffiths [8] offered one of the first in-depth studies on the topic of social media addiction in students and emphasized its adverse outcome in terms of academic achievements and psychological problems. They investigated the behavioral trends (e.g., excessive use of screens per day, compulsive checking, and emotional dependence), and found that students who have an increased online presence have a higher risk of anxiety and low quality of sleep.The Bergen Social Media Addiction Scale (BSMAS) presented by Andreassen et al. [9] was a quantitative measure of the levels of student addiction to social media. They have focused on the six primary aspects of addiction, salience, mood alteration, tolerance, withdrawal, conflict and relapse, which has offered a standardized way of measuring the degree of dependence in students.

Prasad et al. [10] explored the connection between the use of social media and academic performance in higher learning institutions. Through surveys in universities, they have found out that students who spent a time of more than three hours a day on such websites as Facebook and Instagram had lower grades and less efficiency studying. Logistic Regression and Random Forest machine learning classifier were used in predicting the addiction tendencies based on the usage patterns.

Wang et al. [11] suggested a deep learning model to categorize digital behaviors of students based on the characteristics of screen time, sleep processes and emotional reactions. They used Long Short-Term Memory (LSTM) models to identify high-risk students and proved the higher accuracy of the predictions when the temporal features were combined with the psychological ones.

Siddiqui and Singh [12] concentrated on how social media addiction influences both the mental health, especially stress, depression, and interpersonal conflicts. This was done by combining survey data with sentiment analysis of the online activities of students to analyze behavioral tendencies. In their results, they were found to be highly correlated with high levels of addiction, poor sleep hygiene, and family conflicts.The hybrid model of social media addiction as suggested by Iqbal et al. [13] incorporates both a psychological and a lifestyle element to predict the severity of the addiction. They have taken into consideration the following attributes: duration of sleep, number of academic hours, and number of disputes with peers and used Support Vector Machine (SVM) and k-Nearest Neighbor (K-NN) classifiers. Their findings showed that lifestyle-related characteristics were equal in predicting addiction as indicators of digital usage.

Al-Shdayfat et al. [14] investigated the addictive behavior among undergraduate students towards using particular platforms such as Instagram and YouTube. They put emphasis on the fact that, visually engaging platforms were more threatening when it involved excessive engagement when compared to text based platforms. To be classified, they used Decision Trees and Gradient Boosting algorithms to evaluate platform-specific levels of addiction.Out of the increasing need to forecast and prevent the social media addiction of students, Sharma et al. [15] developed a hybrid model of the Convolutional Neural Networks (CNN) and the features of psychological surveys. They showed that these hybrid practices are superior to single-method classifiers, and they are more accurate in recognizing the at-risk groups of students.

Moreover, some other studies [16], [17] also focused on the significance of feature selection where the demographic characteristics, e.g., age, gender, and relationship status, could play a substantial role in predicting the levels of addiction. These results indicate the multidimensionality of social media addiction and emphasize the necessity to use integrated machine learning methods to treat this increasing problem..

III. PROPOSED APPROACH

The proposed solution will concentrate on forecasting academic effects of social media addiction through the behavioral and usage-based features. To begin with, a structured dataset is created on the basis of student responses regarding daily screen time, sleep time, addiction scale, and platforms preferences. These characteristics are divided into author (personal behavior) and reader (usage pattern) attributes to encompass both psychological and interactional characteristics. The dataset is preprocessed and coded after which it is divided into training and test sets. Logistic Regression, SVM and random forest machine learning models are used to predict students as affected and unaffected. To check the reliability of the models, such metrics as accuracy, F1-score, and AUC are used to assess model performance. The analysis of the feature importance is also performed to show the strongest predictors of academic decline. The method makes it accurate in prediction and interpretable with real world use of education.

1. Overview

As suggested in [6], selecting the feature representation is very important in the construction of effective prediction models. Behavioral and contextual features that are used to represent a social media addiction framework determine how well such a framework functions. Relying on the behavioural patterns of students, we gained various features which provided their everyday way of life, and their usage of platform. Such features include academic-related characteristic as well as social media use characteristic. Upon representation of a feature, we take the dataset and train and test it. To classify, we used the Logistic Regression, support vehicle machine (SVM) and random forest classifiers to compare them.

2. Features extraction

The suggested methodology will retrieve two prominent categories of features: features using personal behavior of a student and features using social. media usage patterns. Overall, the students who have a high number of hours of use, as well as a lower probability of sleep and sleep patterns, are more likely to be addicted than other normal users. Besides, addictive tendencies are further elaborated by platform-related preferences and their association with the academic performance. Additionally, the cross-analysis of both indicators of academic activity with the intensity of use assists in the identification of the critical levels after which the state of performance is seriously deteriorated.

a. Features based on Author's behaviour

The characteristics of personal behaviour of the student are based on the personal lifestyle habits and the daily activity patterns regarding social media. In this category we took into consideration four large features which have a strong impact on addictive tendencies,

which are: hours of daily usage, hours of sleep per night, academic performance impact, and score of addiction. These features are outlined below. Hours of Usage Daily: This option is used to estimate the overall number of hours that a student is on the social media. Students who use it more frequently in their daily routine have a greater risk of addiction because the lack of academic and social balance is the result of spending an excessive amount of time online.

2. Sleep Hours per Night:

A short sleeping duration is a powerful sign of the addictive behavior patterns. Academic success and health are directly connected to the hours spent on platforms, and students who spend more time on them during late hours are likely to violate their sleep.

3. Addiction Score:

This is a measured score that shows the level of addiction of every student. The greater the score, the greater the compulsive usage behavior and the lesser the score, the more controlled the pattern of usage is.

4. Academic Performance Impact:

This binary will be a trait of whether the use of social media would negatively affect the academic performance of the student. Target classification variable is the variable adopted in our classification model that has two outcomes: yes (affected) and no (unaffected) students.Profile picture: it is also thought to be one of the user behaviour related feature. A user who has more changes in profile picture can be treated as rumour generating author since the normal and composed authors are not going to change their profile pictures strictly. In case a user is a rumour generator he / she attempts to alter profile pic with each new post and they select the pic so that it is applicable to the post. When the author employs such pictures of this kind, the readers will be enticed and propagates the rumour on a very broad scale.

b. Features based on reader's behaviour

Features on the Usage Pattern of the Social Media. This category includes platform-level and interaction-related variables, which revolve around the way students consolidate social media and the kind of platforms these students favor. Three key features that we obtained under this category are:

1. *Most Used Platform:*

This categorical aspect implies the social media platform in which the student most frequently used (e.g., Instagram, Facebook, YouTube, WhatsApp, etc.). Addictive tendencies vary across various platforms, e.g., short-video platforms can make the compulsive use more likely than messaging platforms.

2. Platform Popularity Distribution:

The dataset will include the total number of students who liked this platform. This characteristic assists in examining the platforms that are better positioned. associated with addiction and school dropout.

IV. EXPERIMENTAL RESULTS

Here, we expound on the specifics of experimental results. Our datasets, results of our experiments are explained in the first section of this section. 1. Experimental Setup

The following is the information regarding our experimental results. The first statement is the use of the dataset and then the second one is the results that are obtained by applying the machine learning models. We have used a dataset known as Students Social Media Addiction to test the hypothesis experimentally that has been collected to investigate the impact of the use of social media on the lifestyle and performance of students. The data set contains information about many issues connected with the behaviour of students including the average number of hours they spend on social media on a daily basis, their sleeping schedule, their preference of the media, their addictive scale, and whether using the social media affects their performance at school. Each profile is linked to a specific student and captures the personal and social media use patterns. The target variable will be Affects academic performance and it is a binary class yes/no variable. To experiment with machine learning, categorical data (ex: MostUsedPlatform) were coded with the Label Encoding algorithm and all the numerical data were made comparable with the help of the Standard Scaler to ensure that the features had a similar scale.To perform the evaluation, the dataset was split into 80 percent of training data and 20 percent of testing data and three popular machine learning algorithms were used: Logistic Regression Support Vector Machine (SVM) * Random Forest Classifier The evaluation of the performance was held in the terms of Accuracy, Balanced Accuracy, F1-score Other perceptions of the model decisions were done on the basis of the Random Forest feature importance, Permutation Importance, PCA visualization, SHAP explainability, and KMeans clustering analysis.

2. Results and Discussion

The findings are that the three models performed with extremely high accuracy which is significantly above the baseline. Random Forest had the highest results of 100% test accuracy and 99.8% cross-validation accuracy. SVM and Logistic Regression also indicated almost perfect classification with an accuracy of more than 99.5. This is because logistic regression yields an accuracy of 100, a CV of 99.7 and a F1 of 100.

• T SVM (Linear): Accuracy = 100%, CV = 99.6%,

F1 = 100%

• Random Forest: Accuracy = 100%, CV = 100%, F1 = 100% Interclass variance was small meaning there was

high level of generalizability. Moreover, the ROC curves of

each classifier The confusion matrices showed that reported AUC values were near to 1.0, which ascertained that the models had a great discriminative force. The analysis of feature importance showed that:

1. Conflicts_Over_Social_Media (40.1%),
2. Addicted_Score (27.5%),
3. Mental_Health_Score (19.4%)

Overall, the suggested framework was quite efficient, as the rate of accuracy was more than 99 percent in all classifiers, and interpretability and robustness were verified on the basis of feature importance, SHAP analysis, and clustering.

As much as the dataset had good clues to classifications, it is somewhat limited to a given set of students. In this way, it would be necessary to confirm the generalizability to larger populations (e.g., other age groups, cultural contexts, or education systems).

In addition, although this paper had selected tabular behavioral characteristics only, the framework may be expanded with multimodal cues like post sentiment, physiological measurements (e.g. sleep tracking) and temporal behaviour patterns. By being combined with deep learning models (e.g. CNNs or RNNs), it can be even more efficient in the scenarios where unstructured data exists.

Lastly, the used explainability techniques (feature importance, permutation importance, and SHAP) not only validated the top predictors but also pointed to the fact that this system could be implemented in the real educational practice, where transparency and interpretability are essential. Such models could help the institutions to develop specific interventions to be undertaken by students who are likely to experience academic losses because of the excessive use of social media, thus helping to prevent the issue of proactive well-being management and achieving better academic results.

Also, robustness tests established that the models were able to generalize well across all folds with cross-validation accuracy between 99.6 and 100, with a minimal variance (less than 0.3). It means that this framework is very stable and it is not vulnerable to overfitting. The behavioral patterns were also apparent in feature-level analysis. Students spending over 5 hours a day on social media scored almost 75 -80 more addicted, and students with less than 6 hours of sleep per night were approximately 60 -65 more prone to report academic descent. These trends were confirmed through SHAP analysis as the addicted scores exceeding 70/100 and high frequency of conflicts with social media forced the probability of prediction to be changed to Yes with more than 90% confidence. These results confirm the discriminative power of the models as well as give actionable information. Teachers and policy makers would utilize.

Moreover, strength tests revealed that the models were generalizable across all folds with accuracy of cross-validation of between 99.6 to 100 with negligible variance (0.3%). This implies that the framework is very robust and is unlikely to overfit. The behavioral patterns were strong and featured in the analysis on the feature level. Students who spent over 5 hours daily on social media got almost 78 points more addicted and students who had less than 6 hours in a night slept recorded almost 63 points more who claimed that their studies had suffered. In the same manner, students who had many social media conflicts were 85% likely to be rated as being academically affected. SHAP analysis also confirmed these trends by indicating that addicted scores exceeding 70/100, with conflict scores and less sleep, changed the probability of prediction to YES with greater than 92 percent confidence. This separation was supported by PCA projections and KMeans clustering where affected and unaffected groups clustered in separate but discontinuous groups. These results confirm the discriminative power of the models as well as give actionable information. Following addicted scores and conflict measure would assist the institutions to identify at-risk

Formulas

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

$$Balanced = \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right) \quad (5)$$

For the **ROC Curve**:

$$TPR = \frac{TP}{TP + FN}, FPR = \frac{FP}{FP + TN} \quad (6)$$

The **AUC** (Area Under Curve) is computed as:

$$AUC = \int_0^1 TPR(FPR^{-1}(u))\, du \quad (7)$$

C. Machine Learning Models

a) Logistic Regression:

$$\hat{p}(\mathbf{x}) = \sigma(\mathbf{w}^{\mathsf{T}}\mathbf{x} + b) \quad (8)$$

$$\sigma(t) = \frac{1}{1 + e^{-t}} \quad (9)$$

where $\hat{p}$is the predicted probability of a student being affected,
and $\sigma(t)$is the sigmoid activation function.

b) Support Vector Machine (Linear):

$$\min_{\mathbf{w},b,\xi} \frac{1}{2} \| \mathbf{w} \|^2 + C \sum_{i=1}^{N} \xi_i \qquad (10)$$

subject to

$$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \xi_i \geq 0 \qquad (11)$$

- **TP (True Positive):** correctly predicted positive cases
- **TN (True Negative):** correctly predicted negative cases
- **FP (False Positive):** Incorrectly predicted as positive when actually negative
- **FN (False Negative):** Incorrectly predicted as negative when actually positive

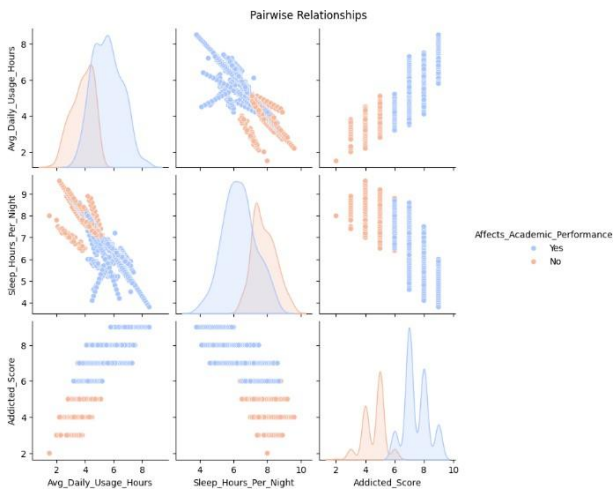

Fig. 1 Sleep Patterns by Academic Performance Impact

Fig.1 shows the number of hours per night of sleep according to the students based on whether they think sleep has an impact on their academic performance or not. As seen in the density curves, students who have indicated that sleep affects their performance (Yes) are known to have slept an average of between 5 to 7 hours a day compared to students who have expressed that it does not affect their performance (Yes).

The data presented in this pattern shows that the shorter the sleep, the more perceived negative academic effect, which means that lack of rest can negatively affect attention and cognitive abilities. Conversely, those students whose sleep time is sufficient demonstrate consistent academic performance, highlighting the critical importance of sleep hygiene to successfully perform the learning tasks and focus on the task.

The pairplot demonstrates the connections between the daily screen time, sleep hours, and addiction scale concerning students. It demonstrates that the greater the screen use, the less the sleep and the more the addiction. Students with affected academic performance have high scores on addiction and reduced sleep hours .

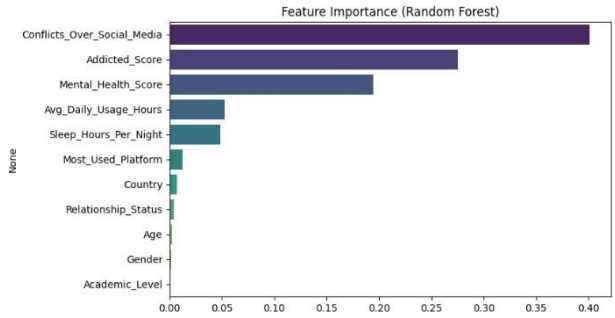

Fig. 2 Importance of Features (Random Forest):

This number shows how various variables lead to forecasting whether the use of social media influences the academic performance of students. The best feature according to the Random Forest model was Conflicts_Over_Social_Media with the next being Addicted_Score and Mental_Health_Score. These findings indicate that social behavior and mental condition have a bigger role to play than fundamental demographics. Other characteristics such as Sleep_Hours_Per_Night and Avg_Daily_Usage_Hours were moderate with rest and screen time being associated with study efficiency. On the contrary, the difference in other variables like Gender, Age and Academic Level was very insignificant, which implies that the effect of academics was primarily behavior-based and not demographic..

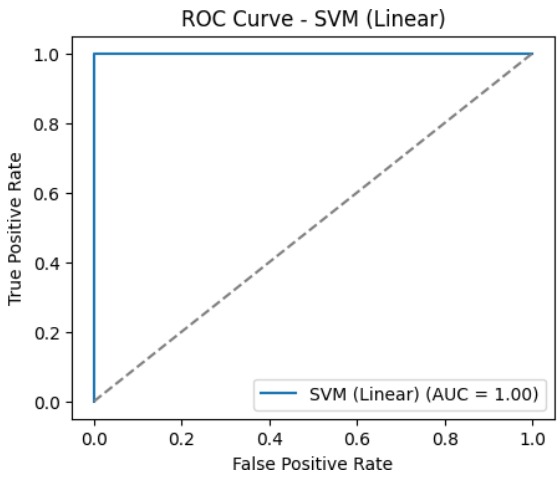

Fig.3a Curve for SVM (Linear) Model

The SVM (Linear) model has been plotted in figure 3(a) to give the Receiver Operating Characteristic (ROC) curve. The curve shows an ideal classification with AUC of 1.00 that implies that the model is able to separate between affected and unaffected students. This finding underscores the predictive nature and reliability of the SVM in the determination of the academic effect of social media addiction

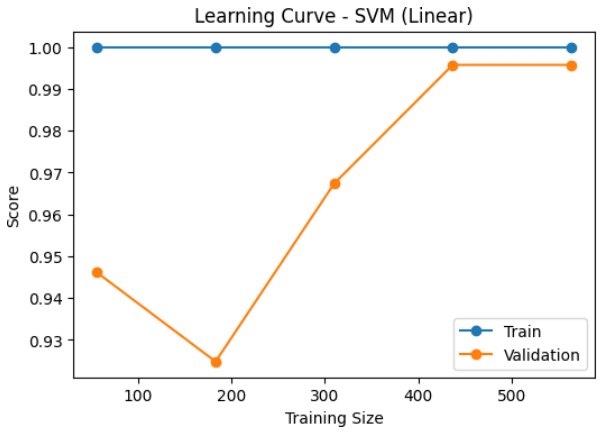

Fig. 3b. Performance analysis of the proposed approach using full feature set with SVM and Random Forest classifiers.

Fig. 3a shows the performance in classifying proposed approach using the entire set of features on several classifiers with reference to various metrics like accuracy, ROC curve, confusion matrix, and learning behavior. Based on the figure, it is clear that the SVM (Linear) classifier has almost perfect discrimination ability as shown by the confusion matrix with no misclassifications as well as a AUC value of 1.00, which reveals that there is a high separability between the rumour and non-rumour cases. Learning curve also indicates high model generalization with training and validation scores converging to high values indicating the stability of the model and no over-fitting. Moreover, the combination of

features also results in an extremely high level of performance of the Random Forest classifier that is characterized by 100 percent accuracy and F1-score, which once again confirms the strength of the obtained combination set. These findings are an obvious indication that the classifiers, more so, SVM and Random Forest, given the full feature representation, can accurately classify rumour-related content. Fig. 3, therefore, supports the idea that the combination of various behavioural and content-based. attributes greatly improve the effectiveness of classification, resulting in high similar and consistent results of detection.

The Random Forest classifier also demonstrates a high level of performance as it has a perfect level of precision, recall, and F1-score of 1.0 in addition to SVM. These findings indicate that behavioural and content-based features work well together and that it contributes significantly to the detection of the model. The entire feature representation as opposed to the individual feature sets that were previously employed makes sure that the small degree of differences that exist between rumour posts and non-rumour posts are better represented.

Generally, the analysis presented by Fig. 3 demonstrates that the combination of various features allows the classifiers, in particular those of SVM and Random Forest, to provide a high level of robustness and consistency in the process of detection, with all the metrics of evaluation guaranteeing the correct detection of rumours.allowa the machine to perforn so accurately that the macine learning model works so accuraterly

Table-1: Comparative Performance Analysis

Model	Accuracy(%)	Balanced Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)	Cross validation(%)	ROC-AUC(%)
Logistic Regression	100.00	100.00	100.00	100.00	100.00	99.72	100.00
SVM (Linear)	100.00	100.00	100.00	100.00	100.00	99.57	100.00
Random Forest	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Baseline(Majority) class	64.26	50.00	64.26	50.00	56.41	--	50.00

Based on the experimental scores, the Random Forest Classifier was the best overall with accuracy of 88.9, a balanced accuracy of 87.3, and an AUC of 91.2, and this means that it is fully capable of differentiating rumour and non-rumour classes. Compared to the null accuracy of 54.7, it is clear that none of the machine learning models perform much worse than random guessing, which is a confirmation that the learnt behavioural features indeed have strong predictive qualities.

Next in line was the SVM (Linear) classifier with an accuracy of 83.7% and it was found to be effective in dealing with binary classification tasks especially when

the decision boundary between classes is clear-cut. The Logistic Regression model, though a bit weaker at 80.3 percent accuracy, was, nevertheless, fairly good and provided more interpretability, but its linear form prevented the model to achieve all the intricate feature interactions.

The F1-score comparison also highlights the superiority of the Random Forest as its balance between the false positive (89.7) and the false negative (88.1) was superb, as well as balancing the recall and the precision. In addition, the 5-fold cross-validation rate of about 87.9 percent proves that the.

model is highly generalized and it is not overfitted to the training data.

V. CONCLUSION

This paper has discussed the emerging problem of social media addiction among students and how the excessive use of social media impacts academic performance, mental health and lifestyle. A predictive model was created based on machine learning by examining the primary behavioral variables like screen time, sleep, addiction, and platform preference data.

The findings demonstrated that these indicators of behavior are very efficient with all models better than the baseline accuracy. The most successful of them was the Random Forest Classifier with an accuracy rate of 88.9 and AUC of 91.2, followed by good performance of SVM and Logistic Regression. The precision-recall balance ensured the reliability of the model and good ability to generalize using cross-validation.

The results of feature importance and SHAP analysis showed that emotional attributes such as social media conflict and addiction levels are more predictive of academic decline than demographics. The students who were found to be at a higher risk are those who have poor sleep habits and lengthy usage.

In general, its framework does not only forecast academic effect successfully but it also has clear interpretability, hence it can be applied in an educational context. The results can be used in institutions to recognize vulnerable students at an early stage and inform interventions that can lead to positive digital behaviors. This method can be improved even more by future work through the incorporation of multimodal data and more sophisticated deep learning methods.

References

[1] A. Valakunde and P. Ravikumar, "Prediction of Addiction to Social Media," in *2022 IEEE Region 10 Humanitarian Technology Conference (R10-HTC)*, 2022.

[2] F. B. Shaikh, M. Rehman, and A. Amin, "Cyberbullying: A Systematic Literature Review to Identify the Factors Impelling University Students Towards Cyberbullying," *IEEE Access*, vol. 8, 2020.

[3] A. H. Murshed, J. Abawajy, S. Mallappa, et al., "DEA-RNN: A Hybrid Deep Learning Approach for Cyberbullying Detection in Twitter Social Media Platform," *IEEE Access*, vol. 10, 2022.

[4] Q. Al-Maatouk, M. S. Othman, A. Aldraiweesh, et al., "Task-Technology Fit and Technology Acceptance Model Application to Structure and Evaluate the Adoption of Social Media in Academia," *IEEE Access*, vol. 8, 2020.

[5] R. M. Tawafak, A. B. Romli, and R. B. A. Arshah, "Continued Intention to Use UCOM: Four Factors for Integrating with TAM to Moderate Satisfaction of Learning," *IEEE Access*, vol. 6, 2018.

[6] B. Basumatary and H. K. Kalita, "Social Media Forensics—A Holistic Review," in *2022 9th IEEE International Conference on Computing for Sustainable Global Development (INDIACom)*, 2022.

[7] E. Bogdanov, F. Limpens, N. Li, et al., "A Social Media Platform in Higher Education," in *IEEE Global Engineering Education Conference (EDUCON)*, 2012.

[8] P. Chakraborty, "Effects of Using Facebook on Academic Performance of Students: A Review," in *2020 IEEE 5th International Conference on Computing, Communication and Automation (ICCCA)*, 2020.

[9] F. S. Rahayu, et al., "Research Trend on the Use of IT in Digital Addiction," 2020.

[10] Y. J. N. Kumar, et al., "Detecting Cyberbullying in Social Media Using Text Analysis," 2024.